

A. Reused Ontologies of FASTO

Table 5
Reused Ontologies and Standards by FASTO

Ontology abbreviation	Ontology name	Version
DDO	Diabetes Diagnosis Ontology	version 1
FMTO (extended version of DMTO)	Diabetes Management Ontology	not reported
RxNorm	RxNorm	not reported
SNOMED CT	Systematized Nomenclature of Medicine-Clinical Terms	not reported
LOINC	Logical Observation Identifiers Names and Codes	not reported
DDI	Drug-Drug Interaction Ontology	not reported
OntoFood	OntoFood	version 1.0
DINTO	Drug-Drug Interactions Ontology	not reported
SMASH	Semantic Mining of Activity, Social, and Health data	not reported
SWE	Sensor Web Enablement	not reported
SWRL TO	SWRL Temporal Ontology	version 3.3
SmartBAN	Smart Body Area Network	not reported
FHIR	Fast Healthcare Interoperability Resources	STU 3
SSN	Semantic Sensor Network	not reported
BFO	Basic Formal Ontology	version 2.0
HL7	Health Level Seven	version 2
VSO	Vitals Signs Ontology	not reported
OAE	Ontology of Adverse Events	not reported

B. ChatGPT Settings

This study used the ChatGPT interface with the GPT 5.5 High model to retain the entire ontology in a single prompt. Each prompt was submitted to the LLM once. Each chat was started with the instruction that no memory or information from previous chats can be used. Regarding the initial ChatGPT user settings, the base style and tone were set to professional, with the characteristic style adjusted to be less warm and less enthusiastic. No role was selected in the settings. Memory was disabled for personalization, and reference chat history was also disabled for record mode.

C. Competency Questions Generated by Task 4

1. Can FASTO distinguish hypoglycemia severity levels, especially level 2 and level 3 events, and use those levels to identify when a diabetes treatment plan should be reevaluated?
2. For patients taking insulin or otherwise at high risk for hypoglycemia, can FASTO represent that glucagon should be prescribed and that family members, caregivers, school personnel, or other support persons should be trained to administer it?
3. Can FASTO classify rescue glucagon formulations by preparation burden and route, such as non-reconstituted glucagon nasal powder, prefilled pen or syringe, dasiglucagon, and powdered glucagon requiring diluent?
4. Can FASTO represent the initial treatment protocol for conscious hypoglycemia: fast-acting glucose or carbohydrate, avoidance of high-fat or high-protein foods for initial treatment, glucose recheck after approximately 15 minutes, and repeated treatment if hypoglycemia persists?
5. Can FASTO represent impaired awareness of hypoglycemia, including screening by validated single-question or multi-item tools such as Pedersen-Bjergaard, Gold, Clarke, or HypoA-Q, and link the result to hypoglycemia risk?

6. Can FASTO represent fear of hypoglycemia as a psychosocial barrier to achieving glycemic targets and well-being, including a screening question and association with anxiety, depression, hypoglycemia confidence, and HbA1c outcomes?
7. Can FASTO represent continuous glucose monitoring metrics such as time in range, time below range, time above range, glucose management indicator, coefficient of variation, and mean glucose, and relate those metrics to individualized glycemic goals?
8. Can FASTO classify diabetes technology into BGM, CGM, insulin pump or continuous subcutaneous insulin infusion, connected insulin pens, and automated insulin delivery systems, including the relation by which AID uses CGM-informed algorithms to modulate insulin delivery?
9. Can FASTO represent advanced hybrid closed-loop system use and its real-world outcome profile, including TIR/TBR changes, severe hypoglycemia, ketoacidosis, skin/adhesive problems, practical barriers, and treatment satisfaction?
10. Can FASTO model mealtime insulin adjustment using carbohydrate intake together with fat and protein intake, concurrent glycemia, glycemic trends, sick-day management, and anticipated physical activity?
11. Can FASTO distinguish insulin-delivery treatment modalities for adults with type 1 diabetes, including continuous subcutaneous insulin infusion, multiple daily injections of basal and prandial insulin, inhaled insulin, and the distinction between insulin analogs and human insulin?
12. Can FASTO determine whether an insulin treatment plan has been reevaluated within a recommended interval, such as every 3 to 6 months, and record the factors used to adjust the plan to individualized glycemic goals?
13. For children and adolescents with type 1 diabetes whose diabetes is partly or wholly managed by someone else, can FASTO link device selection, CGM education, glucagon preparedness, and learning plans to caregiver skills and preferences?
14. Can FASTO represent age- and development-sensitive glycemic assessment frequency, including more frequent assessment during rapid growth and development in children and adolescents and individualized targets for older adults?
15. Can FASTO represent transition planning from pediatric to adult diabetes care, including readiness, transfer timing, and continuity of diabetes-management responsibilities?
16. For older adults with diabetes, can FASTO represent geriatric syndromes and related risks—such as cognitive impairment, depression, falls, frailty, persistent pain, urinary incontinence, hypoglycemia, and polypharmacy—and use them to individualize goals and therapeutic approaches?
17. Can FASTO represent social determinants of health and treatment-burden factors—including financial considerations, food insecurity, support systems, individual values, and preferences—as inputs to treatment decisions and hypoglycemia-risk mitigation?
18. Can FASTO represent dietary patterns and evidence strength for type 1 diabetes management—such as high-fiber diets and carbohydrate-restricted diets—and link them to outcomes including HbA1c, TIR, coefficient of variation, hypoglycemia, insulin dose, and anthropometric measures?

D. Study Limitations

We provided an initial study addressing ontology quality and timeliness assessment within a ChatGPT workflow, showing that the LLM can support ontology maintenance semi-automatically. To provide a workflow that can responsibly guide ontology updates and corrections, a proper validation against independent expert gold standards that yield quantitative metrics, such as precision and recall, is required.

A further limitation of this study is that the ChatGPT interface is not an independent LLM model but rather an extended agentic and tool-based system. Also, its file processing and platform-level behavior may be inconsistent, affecting the output. Instructing the model not to use memory and disabling

memory in the settings may not isolate memory influence. Moreover, ChatGPT does not enable exact reproducibility because inference-level parameters are not exposed, and model parameters cannot be modified (e.g., temperature, random seed, token allocation, and consumption). Consequently, API based or locally deployable LLMs (e.g., Llama 3.3 70b) should be evaluated. The prompts should also be repeatedly run to validate reliability and consistency. Future work should also focus on calibrated uncertainty estimates and a mechanism to detect false mappings or hallucinations. In addition, LLM output was restricted to a JSON schema to increase consistency and prevent hallucination. However, while structured outputs facilitate reproducible data extraction, format constraints may limit LLM performance, especially in reasoning [36]. Hence, future work should conduct ablation studies and explore multiple output structures. Finally, the workflow should be tested with various ontologies across diverse domains.